

TrES-2b

R-2299

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_110520 · created 2026-08-17T22:53:31+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2455701.888506 +/- 8.3e-05 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.126 +/- 0.011
Transit depth (Rp/Rs)^2	1.58 +/- 0.27 [%]
Orbital Inclination (inc)	84.18 +/- 0.39
Airmass coefficient 1 (a1)	1.2382 +/- 0.0082
Airmass coefficient 2 (a2)	-0.2 +/- 0.017
Scatter in the residuals of the lightcurve fit is	0.67 %
Best Comparison Star	None
Optimal Aperture	4.61
Optimal Annulus	9.23
Transit Duration (day)	0.0788 +/- 0.0056

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.126 $\pm$ 0.011 vs 0.1254 (deviation 0.05 σ)
Duration fit vs published	0.0788 d vs 0.0746 d (deviation 0.75 σ)
Mid-time O–C	-0.1 min
Depth SNR	11.3
Residual scatter	0.67 %

Light curve

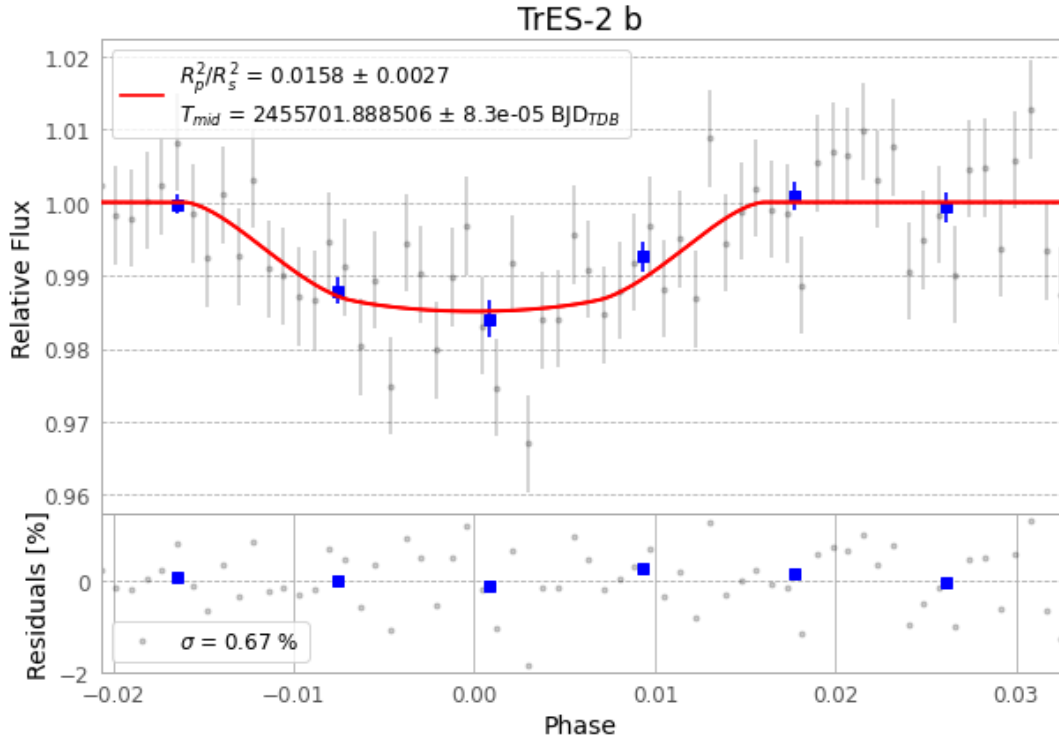


Figure 1 — FinalLightCurve_TrES-2 b_2011-05-20.png (SHA-256 7bdf2a0447decc12...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [354, 239], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 31fad2e9864c2fd1...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

64 FITS frames (42 MB), TRES-2110520080314.FITS → TRES-2110520111213.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-110520.log (5677e38107eb...), FinalLightCurve_TrES-2 b_2011-05-20.png (7bdf2a0447de...), FinalLightCurve_TrES-2 b_2011-05-20.csv (aad30590ad10...)

Compute resources

Wall time 133.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-17 22:53Z.